**Tertiary Infotech Academy Pte Ltd**
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# LESSON PLAN

For

# AI for Lean Manufacturing

TGS Ref No: TGS-2023020425

Conducted by

**Tertiary Infotech Academy Pte Ltd**
UEN: 201200696W

**Version 7.0**

## DOCUMENT VERSION CONTROL RECORD

| Version Number | Effective Date of Release | Summary of Included Changes | Author |
|---|---|---|---|
| 6.0 | Previous release | Legacy two-day lesson plan for the previous course title. | Tertiary Infotech Academy Pte Ltd |
| 7.0 | 19 August 2026 | Renamed course and rebuilt the two-day plan around the final slide map, ten activities, AI-enabled manufacturing content and the original 60+90 minute assessment timings. | Dr Alfred Ang |

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# Course Information

| Field | Details |
|---|---|
| Course title | AI for Lean Manufacturing |
| Course code | TGS-2023020425 |
| TSC | Lean Manufacturing (PRE-OPR-4064-1.1) |
| Duration | 2 days \| 16 instructional/assessment hours \| 9:30 am-6:30 pm daily |
| Assessment | WA 60 minutes + Practical Performance 90 minutes, open book |
| Delivery | Instructor-led concepts, demonstrations, case studies and evidence-based activities |

# Learning Outcomes

- LO1: Identify eight types of production wastes within the organisation, benefits of lean manufacturing and steps to achieve lean performance results.
- LO2: Apply lean principles to compute production capacity and TAKT time.
- LO3: Perform ABC analysis for inventory classification using selective inventory control.
- LO4: Calculate economic batch quantity and number of kanbans required.
- LO5: Identify objectives of production and inventory control and types of inventories by function and condition.
- LO6: Implement lean manufacturing tools for improved inventory flow and material flow.
- LO7: Implement lean manufacturing in an organisation at various levels and within various areas for improvement.
- LO8: Determine the necessity for preventive maintenance within the organisation.
- LO9: Identify performance measurement areas in lean manufacturing and steps required for successful lean implementation.

# Course Schedule

## Day 1

| Time | Minutes | Session | Content and activity |
|---|---|---|---|
| 09:30-10:00 | 30 | Admin | Welcome, digital attendance, introductions, outcomes and assessment briefing |
| 10:00-10:50 | 50 | Topic 1 | Lean purpose, customer value, waste and five principles |
| 10:50-11:00 | 10 | Break | Tea break |

| Time | Minutes | Session | Content and activity |
|---|---|---|---|
| | | | |
| 11:00-13:00 | 120 | Topic 1 / Activities | Waste walk, SIPOC/value stream, capacity and takt (Activities 1-3) |
| 13:00-14:00 | 60 | Lunch | Lunch break |
| 14:00-15:30 | 90 | Topic 2 | Inventory objectives/types, ABC and control policies (Activity 4) |
| 15:30-15:40 | 10 | Break | Tea break |
| 15:40-17:10 | 90 | Topic 2 | EBQ, kanban, JIT, 5S, flow tools and AI inventory support (Activity 5) |
| 17:10-18:20 | 70 | Activity | Root-cause case study using 5 Whys and Fishbone (Activity 6) |
| 18:20-18:30 | 10 | Recap | Day 1 recap and PM digital attendance |

*Day 1 total excluding lunch: 480 minutes (8 hours).*

## Day 2

| Time | Minutes | Session | Content and activity |
|---|---|---|---|
| 09:30-09:45 | 15 | Recap | Day 1 recap and AM digital attendance |
| 09:45-10:45 | 60 | Topic 3 | Preventive maintenance, FMEA, TPM, OEE and metric trees |
| 10:45-10:55 | 10 | Break | Tea break |
| 10:55-12:10 | 75 | Topic 3 / Activities | AI visual quality and predictive maintenance (Activities 7-8) |
| 12:10-13:00 | 50 | Topic 3 | Manufacturing AI, GenAI, data readiness, risk and governance |
| 13:00-14:00 | 60 | Lunch | Lunch break |
| 14:00-15:30 | 90 | Activities | Grounded GenAI and Lean-AI capstone charter (Activities 9-10) |
| 15:30-15:40 | 10 | Break | Tea break |
| 15:40-15:50 | 10 | Recap | Course synthesis and |

| Time | Minutes | Session | Content and activity |
|---|---|---|---|
| | | | evidence check |
| 15:50-16:00 | 10 | Assessment | Briefing for Assessment and assessment digital attendance |
| 16:00-17:00 | 60 | Assessment | Written Assessment (SAQ), open book |
| 17:00-18:30 | 90 | Assessment | Practical Performance, open book |

*Day 2 total excluding lunch: 480 minutes (8 hours).*

## Slide and Activity Alignment

| Topic | Final deck range | Activities | K/A mapping |
|---|---|---|---|
| Topic 1: Lean Foundations, Waste and Flow | Slides 16-48 | Activity 1, Activity 2, Activity 3 | K1 K2 K4 A1 A4 |
| Topic 2: Inventory and Production Control | Slides 50-82 | Activity 4, Activity 5, Activity 6 | K5 K6 K7 A2 A3 |
| Topic 3: AI-enabled Lean Implementation and Maintenance | Slides 84-127 | Activity 7, Activity 8, Activity 9, Activity 10 | K3 K8 A5 A6 |

## Activity Facilitation and Evidence

| Activity | Title | Slides | Guide | Mapping | Evidence |
|---|---|---|---|---|---|
| Activity 1 | Waste Walk and AI Opportunity Scan | Slides 39-41 | 60 min | K1 K2 A1 | Completed waste register, Pareto summary, problem statement and AI/non-AI countermeasure screen. |
| Activity 2 | SIPOC and Value-stream Decision Map | Slides 42-44 | 60 min | K4 A1 | SIPOC, current-state value-stream worksheet, flow/pull improvement hypothesis and evidence plan. |
| Activity 3 | Capacity, Takt and Bottleneck Analysis | Slides 45-47 | 75 min | K4 A4 | Capacity table, takt comparison, bottleneck decision and countermeasure experiment. |
| Activity 4 | ABC Inventory | Slides 73-75 | 75 min | K5 K6 A2 | Ranked ABC |

| Activity | Title | Slides | Guide | Mapping | Evidence |
|---|---|---|---|---|---|
| | Classification with Criticality | | | | table, cumulative-value chart, criticality overrides and class-control policy. |
| Activity 5 | EBQ and Kanban Control Design | Slides 76-78 | 75 min | K7 A3 | EBQ calculation, kanban quantity, sensitivity table, card/bin rule and review trigger. |
| Activity 6 | Root Cause with 5 Whys and Fishbone | Slides 79-81 | 60 min | K7 A6 | Problem statement, 5 Whys chain, fishbone map, evidence plan and countermeasure test. |
| Activity 7 | AI Visual Quality Pilot | Slides 115-117 | 75 min | K3 A6 | Use-case canvas, data plan, confusion-matrix interpretation, decision rights and pilot acceptance criteria. |
| Activity 8 | Predictive-maintenance Decision Design | Slides 118-120 | 75 min | K3 A5 | Criticality/FMEA screen, condition trend, action threshold, work-order recommendation and fallback. |
| Activity 9 | Grounded GenAI for Standard Work | Slides 121-123 | 60 min | K8 A6 | Grounded prompt, source hierarchy, draft instruction, reviewer checklist and controlled-release workflow. |
| Activity 10 | Lean-AI Pilot Charter and Control Plan | Slides 124-126 | 90 min | K1-K8 A1-A6 | A3/pilot charter, SIPOC, baseline, prioritisation matrix, risk controls, ROI, control plan and executive recommendation. |

## Assessment Administration

- Briefing for Assessment appears before assessment administration in both deck and schedule.
- Written Assessment: 8 open-ended questions covering K1-K8; 60 minutes; open book.
- Practical Performance: 6 tasks covering A1-A6; 90 minutes; open book.

- Candidate papers only are linked on the LMS. Answer keys remain trainer-only.
- Assessment digital attendance is completed before the assessment begins.